

Project Topic 1: Analyse E-Commerce Sales to Improve Customer Conversion & Retention (RetailTech)

Context:

An e-commerce retailer aims to improve customer conversion and retention by analyzing behavior across product categories, price points, and customer segments. The curated dataset includes customer profiles, geolocations, products, orders, order items, and reviews. Your role as a data analyst is to explore and analyze this dataset to help the business identify drivers of first purchase conversion, repeat purchases, basket growth, and timely delivery- all feeding into profitable, loyal customer relationships.

Business Objective 1: To identify category- and segment-level drivers of first purchase conversion.

Sample of Business Questions

- Which product categories and subcategories have the highest add-to-cart to purchase conversion and lowest cancellation rates?
- How do conversion rates vary by customer segment (New vs. Loyal), gender, and age group?

Business Objective 2: To understand repeat purchase and loyalty patterns across demographics, categories, and delivery performance.

Sample of Business Questions

- Which categories and brands most strongly predict repeat purchases among New customers transitioning to Loyal?
- How do discount rates, shipping costs, and unit prices affect reorder probability within 30/60/90 days?

Business Objective 3: To optimize pricing, discounting, and shipping for profitable growth.

Sample of Business Questions

- What is item-level margin after discount and shipping, by category, brand, and stock availability?
- How do changes in discount bands influence conversion and average order value across segments?

Business Objective 4: To leverage reviews to enhance conversion and retention.

Sample of Business Questions

- Which categories and brands consistently earn high ratings and low returns/cancellations?
- How do review scores correlate with reorder rates and time-to-next-purchase?